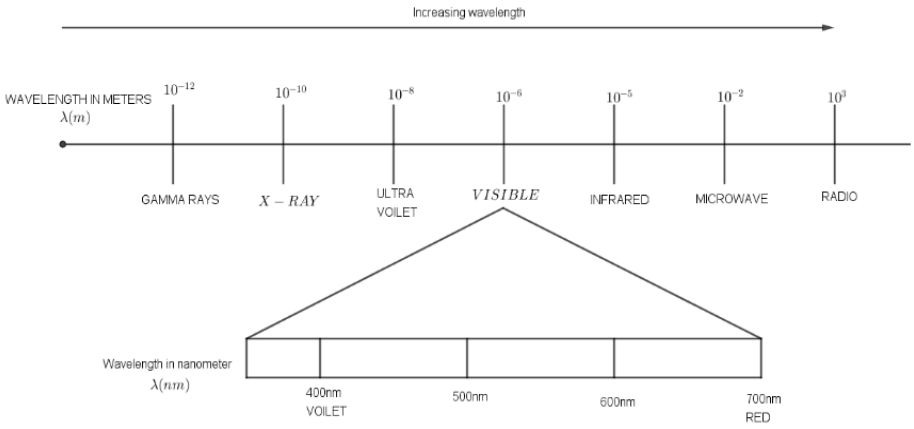
examine internal organs, and monitor the progression of diseases. Techniques such as image enhancement, segmentation, and noise removal make medical images clearer and support more accurate diagnosis and treatment planning.

- **Satellite Imaging and Remote Sensing:** Image processing is used to analyse images captured by satellites, aircraft, and drones to study the Earth's surface and atmosphere. It helps identify land-use patterns, monitor deforestation, assess agricultural conditions, track changes in glaciers, and detect areas affected by natural disasters such as floods and forest fires. These applications support environmental monitoring, urban planning, weather analysis, and disaster management.
- **Industrial Automation and Quality Control:** In manufacturing industries, image processing is used to inspect products and detect defects automatically. Cameras capture images of products, and image processing algorithms analyse them to identify cracks, scratches, incorrect dimensions, surface defects, or missing components. This improves product quality, reduces human error, increases production efficiency, and enables faster inspection on assembly lines.
- **Security and Surveillance:** Image processing plays an important role in security systems by analysing images and video footage to identify people, objects, and suspicious activities. It is used in facial recognition, fingerprint identification, number plate recognition, and motion detection. These techniques help improve access control, support law enforcement investigations, monitor restricted areas, and enhance security at airports, offices, and public places.
- **Agriculture:** Image processing is used in modern agriculture to monitor crop health, detect plant diseases, identify pests, and estimate crop growth. Images captured using drones, satellites, or cameras are analysed to identify changes in plant colour, leaf texture, and vegetation patterns. This helps farmers detect problems at an early stage, apply water and fertilisers more efficiently, reduce crop losses, and improve overall agricultural productivity.

Imaging in the electromagnetic spectrum



1. Radio Waves
    - Longest wavelength and lowest frequency.
    - Used in radio, television, and communication systems.
2. Microwaves
    - Used in microwave ovens, radar, Wi-Fi, and satellite communication.
    - Commonly used for wireless data transmission.
3. Infrared Rays
    - Associated with heat radiation.
    - Used in remote controls, thermal imaging, and night-vision devices.
4. Visible Light
    - The only part of the electromagnetic spectrum visible to the human eye.
    - Used in vision, photography, and illumination.
5. Ultraviolet Rays
    - Have higher frequency than visible light.
    - Used in sterilization, forensic detection, and fluorescence.
6. X-rays
    - Have high energy and can penetrate soft tissues.
    - Used in medical imaging and security scanners.
7. Gamma Rays
    - Shortest wavelength and highest frequency.
    - Produced in nuclear reactions and used in cancer treatment and sterilization.

Basics of Image Processing in Java

1. **BufferedImage**
   BufferedImage is a class in Java's java.awt.image package used to represent and manipulate images stored in memory. It allows programmers to access and modify individual pixels and perform image-processing operations such as grayscale conversion and colour modification.
   Package: java.awt.image.BufferedImage
   Five methods of BufferedImage:
    - **getWidth()** – Returns the width of the image in pixels.
    - **getHeight()** – Returns the height of the image in pixels.
    - **getRGB(int x, int y)** – Returns the colour value of the pixel at the specified coordinates.
    - **setRGB(int x, int y, int rgb)** – Sets the colour value of a pixel at the specified coordinates.
    - **getType()** – Returns the type of the image, indicating how its pixel data is stored.
2. **Color**
   Color is a class in Java's java.awt package used to represent colours. It supports the RGB (Red, Green, Blue) colour model and allows programmers to create colours, retrieve colour components, and manipulate pixel colours during image processing.
   Package: java.awt.Color
   Five methods of Color:
    - **getRed()** – Returns the red component of the colour, with a value ranging from 0 to 255.
    - **getGreen()** – Returns the green component of the colour, with a value ranging from 0 to 255.